

# Ballot words and the Morin Thom polynomials

Chern++

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## Abstract

The Chern coefficients  $C(M)$  of the Morin Thom polynomials  $\text{Tp}_{A_d}$  are indexed by zero-sum multisets  $M$  of  $d$  integers. Let  $b(M)$  be the number of orderings of  $M$  whose proper partial sums are nonnegative. We prove, given two geometric inputs that we state precisely, that  $C(M) = b(M)$  whenever  $\max M \leq 1$ . On that sector  $b(M)$  is a Kreweras number. We conjecture that  $C(M) \geq b(M)$  for every  $M$ , with equality exactly on that sector. This implies Rimányi's positivity conjecture. We prove it, with its equality case, for  $d \leq 4$ . The case  $d = 4$  rests on a new explicit positive product formula for the chamber series together with exact integer-set inclusions. The conjecture also holds on every exact coefficient we can compute: all  $d \leq 7$  on large boxes, including  $\text{Tp}_{A_7}$  to  $\ell = 5$ ,  $\text{Tp}_{A_6}$  to  $\ell = 12$  and every packet of charge  $\leq 7$  at  $d = 7$ .

## 1 Setting

By Rimányi's  $\ell$ -free reduction [2] and the Bérczi–Szenes residue formula [1],

$$\text{Tp}_{A_d}^\ell = \sum_M C(M) \prod_{a \in M} c_{\ell+1+a},$$

where  $M$  ranges over multisets of  $d$  integers summing to 0,  $c_0 = 1$ , and  $c_j = 0$  for  $j < 0$ . The coefficient  $C(M)$  does not depend on  $\ell$ . In chamber coordinates,

$$C(M) = \sum_{\alpha} A_{\beta(\alpha)},$$

where the sum is over the distinct orderings  $\alpha$  of  $M$ , the prefix sums  $\beta_j = \alpha_1 + \dots + \alpha_j$  ( $j < d$ ) must be nonnegative, and  $F_d = \sum_{\beta} A_{\beta} x^{\beta}$  is the Morin chamber series. Rimányi's conjecture is  $C(M) \geq 0$ .

**Definition 1.** A *ballot ordering* of  $M$  is an ordering whose proper prefix sums are all nonnegative. Let  $b(M)$  be the number of distinct ballot orderings. Equivalently,  $b(M)$  is  $C(M)$  computed with the pure chain series  $\prod_{j=1}^{d-1} (1 - x_j)^{-1}$ , all of whose coefficients are 1, in place of  $F_d$ .

The decreasing ordering is always ballot, so  $b(M) \geq 1$ . The *charge* of  $M$  is  $\max M$ . The *plane sector* is charge  $\leq 1$ ; for  $\ell = L - 1$  these are the Chern monomials whose largest index is at most  $\ell + 2$ .

## 2 The plane sector

**Theorem 2.** Assume the geometric inputs (I1) and (I3) of §2.1. If  $\max M \leq 1$  then  $C(M) = b(M)$ . Write  $M = (1^{d-b}, 1 - s_1, \dots, 1 - s_b)$  with every  $s_i \geq 1$ , and let  $m_i$  be the number of  $s$ 's equal to  $i$ . Then this common value is the number of noncrossing partitions of  $[d]$  with block sizes  $s_1, \dots, s_b$ :

$$C(M) = b(M) = \frac{d!}{(d - b + 1)! \prod_i m_i!}.$$

Independently of (I1) and (I3), the identity  $C(M) = b(M)$  is verified by exact computation from the chamber series on every plane-sector packet with  $d \leq 7$ .

## 2.1 The geometric inputs

Fix  $n = d + 1$  and  $r \geq 2$ .

- Let  $C_{n,r}$  be the *curvilinear component* of the punctual Hilbert scheme  $\text{Hilb}_0^n(\mathbb{C}^r)$ . This is the closure of the locus of length- $n$  subschemes supported at the origin and isomorphic to  $\text{Spec } \mathbb{C}[t]/(t^n)$ . It is irreducible of dimension  $(r - 1)d$ .
- Let  $V$  be the tautological rank- $n$  bundle, whose fibre at  $Z$  is  $\mathcal{O}_Z$ .
- Let  $E = \ker(V \rightarrow \mathcal{O})$  be its rank- $d$  augmentation part, with Chern roots  $\epsilon_1, \dots, \epsilon_d$ .

Write the Thom polynomial in the Chern classes  $c_i = e_i(x_1, \dots, x_q)$  of formal roots  $x_1, \dots, x_q$ , where  $q = r + \ell$ ; taking  $r$  large enough makes  $q$  exceed every index that occurs.

$$(I1) \text{ Incidence formula. } \text{Tp}_{A_d}^\ell = \int_{C_{n,r}} \prod_{a=1}^q \prod_{i=1}^d (x_a + \epsilon_i).$$

This is the curvilinear form of the Bérczi–Szenes theorem [1]. Reason: a  $q$ -tuple of function germs on  $\mathbb{C}^r$  defines an  $A_d$  singularity exactly when its zero scheme is curvilinear of length  $n$ . The incidence variety of pairs (curvilinear  $Z$ ,  $q$ -tuple vanishing on  $Z$ ) is a vector bundle over  $C_{n,r}$ , namely the kernel of evaluation onto  $E \otimes \mathbb{C}^q$ . Its projection to the space of  $q$ -tuples is generically one-to-one onto the  $A_d$  locus, which has codimension  $qd - (r - 1)d = d(\ell + 1)$ . The pushforward is the displayed integral. In exactly this form, with this proof, it is stated in [3].

- (I2) *Extraction.* Let  $\lambda = (\lambda_1, \dots, \lambda_s)$  have all parts  $\leq q$  and  $s \leq d$ , and let  $\nu = \text{sort}(q^{d-s}, q - \lambda_1, \dots, q - \lambda_s)$ . Then the coefficient of the Chern monomial  $c_\lambda$  in (I1) is  $\int_{C_{n,r}} m_\nu(\epsilon)$ , where  $m_\nu$  is the monomial symmetric function.

*Proof.*  $\prod_i \prod_a (x_a + \epsilon_i) = \prod_i \sum_{j=0}^q e_j(x) \epsilon_i^{q-j}$ . The coefficient of  $e_{\lambda_1}(x) \cdots e_{\lambda_s}(x) e_0(x)^{d-s}$  is the sum over distinct assignments of the indices  $\lambda_1, \dots, \lambda_s, 0^{d-s}$  to the  $d$  roots, that is,  $m_\nu(\epsilon)$ .  $\square$

- (I3) *Hyperplane lemma.*  $c_d(E) \cap [C_{n,r}] = [C_{n,r-1}]$ , where  $C_{n,r-1}$  is embedded as the subschemes lying in a coordinate hyperplane.

Reason: the last coordinate  $u_r$  gives a section of  $E$ . It vanishes exactly on the subschemes contained in  $\{u_r = 0\}$ . On the standard curvilinear chart  $I = (u_2 - p_2(t), \dots, u_r - p_r(t), t^n)$  it is the regular  $d$ -tuple of coefficients of  $p_r$ , so the zero locus is reduced of the expected codimension  $d$ . This is stated and proved in this form in [3].

- (I4) *The plane* (classical). For  $r = 2$ ,  $C_{n,2} = \text{Hilb}_0^n(\mathbb{C}^2)$  by Briançon’s theorem [4]. Its cohomology is identified by Lehn–Sorger [5] with the associated graded of the class algebra of  $S_n$  (filtered by  $n$  minus the number of cycles). Under this identification the elementary symmetric functions of the  $\epsilon_i$  go to those of the Jucys–Murphy elements  $J_k = \sum_{a < k} (a \ k)$ . The top degree is spanned by the class of the  $n$ -cycles, normalised by  $\int c_d(E) = 1$ . So for  $|\nu| = d$ ,  $\int_{C_{n,2}} m_\nu(\epsilon) = [\sigma] m_\nu(J_1, \dots, J_n)$  for any  $n$ -cycle  $\sigma$ .

**Reduction to the plane.** Take  $\ell = L - 1$  and a plane-sector  $M$ , so the Chern monomial is  $c_\lambda$  with  $\lambda_i = L + \alpha_i \leq L + 1$  for the entries  $\alpha_i$  of  $M$  with  $L + \alpha_i > 0$ . With  $q = r + L - 1$ , every entry of  $\nu$  is at least  $q - L - 1 = r - 2$ . While  $r > 2$  all entries are positive, so  $m_\nu(\epsilon) = (\epsilon_1 \cdots \epsilon_d) m_{\nu-1}(\epsilon) = c_d(E) m_{\nu-1}(\epsilon)$ . By (I3) this lowers  $r$  by one and every entry of

$\nu$  by one. At  $r = 2$ ,  $q = L + 1$  and the entries of  $\nu$  are  $L + 1 - \lambda_i = 1 - \alpha_i$ , together with  $q = 1 - (-L)$  for the entries  $\alpha = -L$  that give  $c_0$ . So  $\nu$  is the multiset  $1 - M$ , and by (I4)

$$C(M) = [\sigma] m_{1-M}(J_1, \dots, J_n).$$

Everything below is new.

**Lemma 3.** *Let  $\sigma(x) = x + 1 \pmod{n}$ . A minimal monotone factorisation of  $\sigma$  is an expression  $\sigma = (a_1 b_1)(a_2 b_2) \cdots (a_{n-1} b_{n-1})$  with  $a_i < b_i$  and  $b_1 \leq \cdots \leq b_{n-1}$ . Its exponent vector is  $e_k = \#\{i : b_i = k\}$ , with  $e_1 = 0$ . Then for every  $e \in \mathbb{Z}_{\geq 0}^n$  with  $\sum e_k = n - 1$ :*

$$\#\{\text{minimal monotone factorisations of } \sigma \text{ with exponent vector } e\} = \begin{cases} 1 & \text{if } D(j) := j - 1 - \sum_{k \leq j} e_k \geq 0 \text{ for all } j \\ 0 & \text{otherwise.} \end{cases}$$

*Proof. Necessity.* The  $n - 1$  transpositions of a minimal factorisation of an  $n$ -cycle are the edges of a tree on  $[n]$ . Those with  $b_i \leq j$  are edges of a forest on  $[j]$ , so there are at most  $j - 1$  of them.

*Existence and uniqueness,* by induction on  $n$ . The case  $n = 1$  is trivial. Let  $m = e_n$  and let  $(a_1 n) \cdots (a_m n)$  be the final  $m$  factors, with product  $C$ . The remaining factors form a minimal monotone factorisation of  $F = \sigma C^{-1}$ , a permutation of  $[n - 1]$  with exactly  $m$  cycles. Since  $C^{-1} = (n a_1 a_2 \cdots a_m)$  we have  $a_1 = \sigma^{-1}(n) = n - 1$  and

$$F(x) = x + 1 \ (x \notin \{a_i\}), \quad F(a_i) = a_{i+1} + 1 \ (i < m), \quad F(a_m) = 1.$$

Because  $F \cdot C = \sigma$  is one cycle and  $C$  is an  $(m + 1)$ -cycle, each cycle of  $F$  contains exactly one  $a_i$ . Starting at  $a_{i+1} + 1$  and stepping by  $+1$ , the cycle through  $a_i$  must meet  $a_i$  before any other  $a_j$ . So  $a_{i+1} < a_i$  with no  $a_j$  in between, and  $a_m$  is the least. Hence  $a_1 > a_2 > \cdots > a_m$ , and  $F$  is the product of the increasing cycles on the consecutive intervals

$$I_m = [1, a_m], \quad I_{m-1} = [a_m + 1, a_{m-1}], \quad \dots, \quad I_1 = [a_2 + 1, n - 1].$$

Transpositions on disjoint intervals commute, so monotone factorisations of  $F$  are exactly shuffles, merged by  $b$ , of monotone factorisations of the interval cycles. Minimality on each interval says the exponents inside  $I_{m-i}$  sum to  $|I_{m-i}| - 1$ . Since  $D$  changes by  $1 - e_k \leq 1$  per step, this means  $D(a_{m-i}) = i$ , and the ballot condition inside the interval means  $D > i$  strictly after  $a_{m-i}$ , up to the next cut. So  $a_{m-i}$  is the last  $j$  with  $D(j) = i$ . When  $e$  is ballot these last visits exist, are increasing in  $i$ , and end at  $D(n - 1) = m - 1$ . So the cut points exist and are forced, and the induction applies to each interval (each is again an increasing cycle, of smaller length).  $\square$

*Proof of Theorem 2.* Expanding  $m_\nu(J_1, \dots, J_n) = \sum_e \prod_k J_k^{e_k}$  over distinct rearrangements  $e$  of  $\nu$  (padded by zeros,  $e_1 = 0$  since  $J_1 = 0$ ), the coefficient of  $\sigma$  counts pairs  $(e, \text{minimal monotone factorisation with exponent vector } e)$ . By Lemma 3 this is the number of ballot vectors  $e$ . Setting  $\alpha_k = 1 - e_{k+1}$  ( $k = 1, \dots, d$ ) turns the condition  $D \geq 0$  into  $\alpha_1 + \cdots + \alpha_j \geq 0$ , and  $e$  into an ordering of  $M$ . So  $C(M) = b(M)$ . For the product formula, apply the cycle lemma to  $-M$  with a final  $-1$  appended:  $n$  integers  $\geq -1$  summing to  $-1$ , exactly one cyclic rotation of which has all proper prefix sums nonnegative. Reversal identifies these rotations with ballot orderings of  $M$ , so  $b(M) = \frac{1}{n} \cdot \frac{n!}{(d-b+1)! \prod_i m_i!}$ . This is Kreweras' count [6].  $\square$

Lemma 3 is checked exhaustively for  $n \leq 7$ ; the totals are the Catalan numbers 1, 2, 5, 14, 42, 132. The identity  $C(M) = b(M)$  is checked on every plane-sector packet for  $d \leq 7$  directly from the chamber series. Both checks are in tier 12 of the test suite.

*Remark 4.* On the plane sector the Morin chamber series is therefore indistinguishable, packet by packet, from the pure chain series  $\prod_j (1 - x_j)^{-1}$ . For example, at  $d = 7$  the packet  $M = (1, 1, 1, 0, -1, -1, -1)$  has 35 ballot orderings. Eight of them carry nonzero  $A_\beta$ , namely 1, 4,  $-1$ ,  $-4$ , 5, 6, 6, 18, and these sum to 35.

### 3 The ballot conjecture

**Conjecture 5** (ballot). *For every zero-sum multiset  $M$  of  $d$  integers,  $C(M) \geq b(M)$ , with equality if and only if  $\max M \leq 1$ . Equivalently: the series  $F_d - \prod_j (1 - x_j)^{-1}$  has nonnegative packet sums, and they vanish exactly on the plane sector.*

**Corollary 6.** *Conjecture 5 implies Rimányi’s conjecture, and more: every Chern coefficient of  $\text{Tp}_{A_d}$  is at least 1 whenever the monomial occurs.*

This explains a feature of the data that had no explanation: across  $d = 4, \dots, 7$  the least Chern coefficient is 1, never 0. It is attained exactly at  $M = 0$  and  $M = (1^{d-1}, 1 - d)$ , the two multisets with  $b(M) = 1$  on the plane sector.

Table 1: The ballot conjecture on every exact table. A violation is  $C(M) < b(M)$ ; the equality columns test that  $C = b$  holds exactly on the plane sector.

| $d$ | box            | packets | violations | plane $C \neq b$ | off-plane $C = b$ | $\min_{\max M \geq 2} C/b$ |
|-----|----------------|---------|------------|------------------|-------------------|----------------------------|
| 2   | level box L=14 | 15      | 0          | 0                | 0                 | 2                          |
| 3   | level box L=14 | 169     | 0          | 0                | 0                 | 2                          |
| 4   | level box L=12 | 1033    | 0          | 0                | 0                 | 9/4                        |
| 5   | level box L=12 | 7166    | 0          | 0                | 0                 | 12/5                       |
| 6   | level box L=11 | 29941   | 0          | 0                | 0                 | 79/30                      |
| 7   | level box L=6  | 8033    | 0          | 0                | 0                 | 58/21                      |
| 7   | charge box p=6 | 8033    | 0          | 0                | 0                 | 58/21                      |
| 6   | level L=13     | 62239   | 0          | 0                | 0                 | 79/30                      |
| 7   | charge p=7     | 16475   | 0          | 0                | 0                 | 58/21                      |

**Evidence.** Table 1 tests the conjecture on every exact table available. The data come from the exact sweep of `chernpp.bboxes`, reconstructed by CRT and verified against a held-back prime. The level box  $\min M \geq -L$  contains every packet of  $\text{Tp}^\ell$  with  $\ell < L$ ; the charge box contains every packet with  $\max M \leq p$ . There is no violation, no plane-sector packet with  $C \neq b$ , and no off-plane packet with  $C = b$ . Off the plane sector  $C(M)/b(M) \geq 2$  throughout. The minimum is 2 at  $(2, -1, -1)$  and grows slowly with  $d$  (9/4, 12/5, 79/30, 58/21), attained on the family  $(2, 1^{d-4}, -1^{d-2})$  and its neighbours.

#### What does not work.

- *Factorial weights.*  $C(p, -1^p) = p!$ , which suggests weighting each ballot word by  $\prod_{\alpha_i > 0} \alpha_i!$ . That bound fails already at  $d = 3$ .
- *Fusion.* Replacing a 2 by two 1s and counting plane-sector ballot words of the longer multiset gives some families exactly. For example,  $C(2, 0^{d-2}, -2) = \binom{d+1}{3} + \binom{d}{2}$ , where both terms are plane-sector counts. But the residual is negative elsewhere.
- *Local step weights.* Keep weight 1 on steps  $\leq 1$  and give each step of size 2 a weight depending on local data: its height, together with the neighbouring steps, the number of remaining steps, or the next return. Fitted exactly on the packets with one 2, every such model that is consistent within a single  $d$  is inconsistent across  $d$ .
- *Formal Jucys–Murphy extension.* Theorem 2 reads  $C(M) = \sum_\alpha [\sigma] \prod_k J_{k+1}^{1-\alpha_k}$  over ballot  $\alpha$ . At charge 2 this needs  $J_k^{-1}$ , which exists for  $k = 2, 3$  (their eigenvalues, the contents of boxes 2 and 3, are nonzero). Where it can be evaluated, it returns the wrong values, some of them fractional ( $C(2, 0, -2) = 7$  against  $23/2$ ). At  $d = 4$  the needed inverses do not exist.

- *Individual coefficients.* No bound on single  $A_\beta$  can work: they are unbounded below ( $-3622$  at  $d = 7$ ), and the smallest symmetrisation that clears all negatives grows with  $d$  [7].

## 4 The conjecture for $d \leq 4$

At fixed  $d$  the plane sector is finite: its entries lie in  $[1-d, 1]$ . So the equality part of Conjecture 5 at fixed  $d$  is a finite check. It needs neither Theorem 2 nor its geometric inputs. The content is the inequality over infinitely many  $M$ , together with strictness off the plane sector. We prove both for  $d \leq 3$ , directly from the chamber series.

In chamber coordinates write  $x = x_1$ ,  $y = x_2$ . The chamber series is a power series in these variables, and its exponents  $\beta \geq 0$  are exactly the prefix-sum vectors of ballot orderings. For  $d \leq 3$  the multidegree numerator is trivial, and the Bérczi–Szenes formula [1] gives

$$F_2 = \frac{1-x}{1-2x}, \quad F_3 = \frac{(1-x)(1-y)(1-xy)}{(1-2x)(1-2xy)(1-y-xy)}.$$

These are the stored chamber algebras of this project, checked in tier 15. The factorisation of the numerator of  $F_3$  is a direct expansion.

**Theorem 7.** *For  $d \leq 3$  and every zero-sum  $M$ :  $C(M) \geq b(M)$ , with equality if and only if  $\max M \leq 1$ . Moreover  $C(M) \geq b(M) + 1$  whenever  $\max M \geq 2$ .*

*Proof.*  $d = 2$ .  $F_2 = 1 + x/(1-2x)$ , so  $C(k, -k) = 2^{k-1}$  for  $k \geq 1$  and  $C(0, 0) = 1$ , while  $b = 1$  throughout.

$d = 3$ . Each factor of  $F_3$  has the form  $\frac{1-u}{1-u-v} = 1 + \frac{v}{1-u-v}$  with  $u, v$  monomials with nonnegative coefficients:

$$F_3 = (1+U)(1+V)(1+W), \quad U = \frac{x}{1-2x}, \quad V = \frac{xy}{1-2xy}, \quad W = \frac{xy}{1-y-xy}.$$

All of  $U, V, W$  have nonnegative coefficients, so  $F_3 \geq 1 + U + V + W + UV + UW$  coefficientwise. Compare with

$$Z_3 = 1 + \frac{x}{(1-x)(1-y)} + \frac{xy}{1-xy},$$

whose coefficient at  $x^i y^j$  is 1 at  $(0, 0)$ , 0 for  $i = 0 < j$ , and  $1 + [i = j]$  for  $i \geq 1$ .

(i)  $V \geq xy/(1-xy)$ , since  $2^{k-1} \geq 1$ .

(ii)  $U + W + UW \geq x/((1-x)(1-y))$ . Write  $[x^i y^j]$  for a coefficient. Then  $[x^i]U = 2^{i-1}$  covers  $j = 0$ .  $W = \sum_{m \geq 0} xy(y+xy)^m$  has  $[x^{1+k}y^{1+m}]W = \binom{m}{k}$ , which covers  $1 \leq i \leq j$ . And  $[x^i y^j]UW \geq [x^{i-1}]U \cdot [xy^j]W = 2^{i-2}$  covers  $1 \leq j < i$ .

So  $F_3 \geq Z_3$  coefficientwise. Next, the packet sums of  $Z_3$  are  $b(M)$ . A ballot ordering  $\alpha$  of  $M = (a, b, c)$  has  $\beta = (\alpha_1, \alpha_1 + \alpha_2)$ . - It lands on the zero set  $\{i = 0 < j\}$  exactly when  $\alpha = (0, k, -k)$ , which requires  $M = (k, 0, -k)$ . - It lands on the diagonal  $i = j \geq 1$  exactly when  $\alpha = (k, 0, -k)$ , with the same  $M$ .

Both occur exactly for  $M = (k, 0, -k)$ ,  $k \geq 1$ , and then they cancel. So the  $Z_3$ -packet sum is the number of ballot orderings, and  $C(M) \geq b(M)$ .

For strictness, every cell with  $i \geq 2$  receives at least one unit beyond what was used to dominate  $Z_3$ .

- $j = 0$ :  $[x^i]U = 2^{i-1} \geq 2$ .
- $1 \leq j < i$ : the unused product  $UV$  gives  $[x^{i-j}]U \cdot [(xy)^j]V \geq 1$ .
- $j \geq i$ : (ii) used only  $W$  there. The term of  $UW$  with  $x$  taken from  $U$ , namely  $[x]U \cdot [x^{i-1}y^j]W = \binom{j-1}{i-2} \geq 1$ , is unused.

If  $\max M \geq 2$ , the decreasing ordering of  $M$  is ballot with  $\beta_1 = \max M \geq 2$ . So  $C(M) \geq b(M) + 1$ . Finally, the plane sector at  $d = 3$  is  $\{(0, 0, 0), (1, 0, -1), (1, 1, -2)\}$ , where  $C = 1, 3, 1 = b$ .  $\square$

#### 4.1 An explicit positive formula for $F_4$

The same shape of argument needs a manifestly positive expression for the chamber series. At  $d = 4$  one exists. In chamber variables  $x_1, x_2, x_3$  put  $s = x_2x_3$  and  $t = x_1x_2x_3$ . The Morin chamber series is

$$F_4 = \frac{N_4}{(1-2x_1)(1-2x_1x_2)(1-x_2-x_1x_2)(1-2t)(1-s-t)(1-x_3-t)(1-2s)},$$

and its numerator factors completely:

$$N_4 = (1-x_1)(1-x_2)(1-x_1x_2)(1-x_3)(1-s)(1-t)(1-s-2t).$$

**Lemma 8.**

$$F_4 = \left( \frac{1}{2} \frac{1}{1-2x_1} + \frac{1}{2} \frac{1}{1-2s} + \frac{x_1^2 s}{(1-2x_1)(1-s-t)} \right) \cdot \frac{1-x_1x_2}{1-2x_1x_2} \cdot \frac{1-x_2}{1-x_2-x_1x_2} \cdot \frac{1-t}{1-2t} \cdot \frac{1-x_3}{1-x_3-t}.$$

Each of the last four factors is  $1 + v/(1-u-v)$  with  $u, v$  nonnegative, and the first factor is a sum of nonnegative series. In particular  $F_4$  has nonnegative coefficients (strong positivity at  $d = 4$ ).

*Proof.* Group the numerator factors  $(1-x_1)(1-s)(1-s-2t)$  with the denominator factors  $(1-2x_1)(1-s-t)(1-2s)$ , and write  $a = x_1$ , so that  $t = as$ . Then

$$(1-a)(1-s)(1-s-2as) = a^2s(1-2s) + \frac{1}{2}(1-2a)(1-s-as) + \frac{1}{2}(1-s-as)(1-2s),$$

which is checked by expanding both sides. Dividing by  $(1-2a)(1-s-as)(1-2s)$  gives the first factor, and the remaining four numerator factors pair with the remaining four denominator factors as displayed.  $\square$

This expression was found by a small exact certificate search, and both identities are pinned in tier 15.

**Theorem 9.** The conclusion of Theorem 7 holds for  $d = 4$ :  $C(M) \geq b(M)$  for every zero-sum  $M$ , with equality if and only if  $\max M \leq 1$ .

*Proof. Step 1: a lower bound for  $F_4$  by supports.* Since  $\frac{1}{2} \frac{1}{1-2y} = \frac{1}{2} + \frac{y}{1-2y}$ , Lemma 8 reads  $F_4 = (1 + X + S + Y) \prod_{i=1}^4 (1 + V_i)$  with

$$X = \frac{x_1}{1-2x_1}, S = \frac{s}{1-2s}, Y = \frac{x_1^2 s}{(1-2x_1)(1-s-t)}, V_1 = \frac{x_1x_2}{1-2x_1x_2}, V_2 = \frac{x_1x_2}{1-x_2-x_1x_2}, V_3 = \frac{t}{1-2t}, V_4 = \frac{t}{1-x_3-t}.$$

Each factor has nonnegative integer coefficients, positive exactly on the following supports. Write  $(a, b, c)$  for an exponent vector. -  $X$ :  $b = c = 0 < a$ . -  $S$ :  $a = 0 < b = c$ . -  $Y$ :  $a \geq 2, b = c \geq 1$ . -  $V_1$ :  $a = b \geq 1, c = 0$ . -  $V_2$ :  $1 \leq a \leq b, c = 0$ . -  $V_3$ :  $a = b = c \geq 1$ . -  $V_4$ :  $1 \leq a = b \leq c$ .

The coefficients are powers of 2 or binomial coefficients  $\binom{m}{j}$  with  $0 \leq j \leq m$ . For example,  $V_2 = \sum_m x_1x_2(x_2 + x_1x_2)^m$  has coefficient  $\binom{m}{j}$  at  $(1+j, 1+m, 0)$ . Expanding the product,  $F_4$  is a sum of 64 series with nonnegative integer coefficients, each positive exactly on a Minkowski sum of these supports. Hence  $F_4(\beta) \geq N(\beta)$ , the number of those 64 supports that contain  $\beta$ .

*Step 2: the comparison series.* The zero set of  $F_4$  is  $\{a = 0, b \neq c\} \cup \{b = 0 < a, c\}$ . The injective map in the text, extended by  $(0, 0, c, -c) \mapsto (c, 0, 0, -c)$  and  $(0, b, -b, 0) \mapsto (b, 0, -b, 0)$ , carries the ballot words landing there bijectively onto the words landing on

$$T = \{b = c \geq 1 \leq a\} \cup \{a = b \geq 1, c = 0\} \cup \{b = a + c, a, c \geq 1\},$$

and every step permutes the entries of  $\alpha$ . So  $Z_4 = P_4 - 1_{\text{zero set}} + 1_T$ , where  $P_4 = \prod_j (1 - x_j)^{-1}$ , has packet sums exactly  $b(M)$ . Its coefficient is  $w = 1_{W_1} + 1_T$ , where  $W_1$  is the complement of the zero set.

*Step 3: exact inclusions.* Let  $D$  be the set of cells whose word is weakly decreasing with  $\alpha_1 \geq 2$ . Every packet with  $\max M \geq 2$  has its decreasing ordering in  $D$ . The functions  $N$  and  $w + 1_D \leq 3$  are defined by Presburger formulas. The inequality  $N \geq w + 1_D$  is equivalent to three inclusions: for  $k = 1, 2, 3$ ,  $\{w + 1_D \geq k\}$  lies inside the set of cells covered by at least  $k$  of the 64 supports. These inclusions, and  $T, D \subseteq W_1$ , are decided exactly by the integer-set library `isl (chernpp.ballot.verify_d4`, one second), with no truncation.

Summing over a packet:  $C(M) \geq \sum N \geq \sum (w + 1_D) = b(M) + [\max M \geq 2]$ . The plane sector at  $d = 4$  consists of five packets, where  $C = b$  is checked directly.  $\square$

## 4.2 Toward $d = 5$

Strong positivity fails for the canonical  $F_5$ , so the first step needs a different numerator with the same packet sums. A polynomial  $K$  in  $z_1, \dots, z_5$  is *residue-null* if replacing the multidegree numerator  $\mathcal{Q}_5$  by  $\mathcal{Q}_5 - K$  leaves every Chern coefficient unchanged. The product

$$K = (2z_1 + z_2 - z_5)(2z_1 - z_2)(z_1 + z_4 - z_5)$$

is such a kernel. Its nullity is proved in [3] by two contour exchanges in the iterated residue, and here it is verified exactly: every packet sum is unchanged at truncations 16, 18, 20 (tier 9).

**Observation 10** (proved, as an identity of rational functions). *Write  $p = x_3x_4$ ,  $q = x_2x_3x_4$  and  $t = x_1x_2x_3x_4$ . The chamber numerator of  $\mathcal{Q}_5 - K$  factors completely:*

$$(1-x_1)(1-x_2)(1-x_1x_2)(1-x_3)(1-x_2x_3)(1-2x_2x_3)(1-x_1x_2x_3)(1-x_4)(1-p)(1-q)(1-t)(1-p-2t)(1-q-2t).$$

*The factor  $(1 - 2x_2x_3)$  cancels a denominator factor. Six of the remaining factors pair with six denominator factors into ratios  $1 + v/(1 - u - v)$ . What is left is the block*

$$\frac{(1-x_1)(1-x_1x_2)(1-p)(1-q)(1-p-2t)(1-q-2t)}{(1-2x_1)(1-2x_1x_2)(1-q-t)(1-p-t)(1-2q)(1-p-q)},$$

*whose numerator is exactly the product of the numerators of two copies of the  $d = 4$  block of Lemma 8. An exact certificate writes the block numerator as  $\sum_S P_S \prod_{r \in S} (1 - f_r) + P_\emptyset$ . It has 17 nonnegative rational parts  $P_S$  over subsets  $S$  of at most five factors, and a nonnegative remainder. Dividing by the block denominator exhibits the block as a nonnegative series. Hence the gauged series  $F_5^K$  has an explicit nonnegative expression. The certificate is stored in `results/d5_block_certificate.json` and re-verified in exact rationals by `chernpp.ballot.verify_d5_block` (tier 15).*

This gives a new, fully factorised route to strong positivity of a residue-equivalent representative of  $F_5$ . It does not yet give Theorem 9 at  $d = 5$ , for two reasons.

- Step 1 counts supports, which needs each term of the expansion to contribute whole units. The certificate's weights are fractional  $(1/12, 5/18, \dots)$ , and no integer certificate of the same shape exists with parts of degree  $\leq 10$ : the integer program is infeasible while its linear relaxation is feasible.
- The argument would rest on the nullity of  $K$ , which is proved only in [3].

A weighted version of Step 3, or an integer certificate of higher degree, would close the first gap.

## 5 Zero insertion

Appending a 0 to  $M$  adds a singleton block and multiplies the Chern monomial by  $c_{\ell+1}$ . For a zero-free base  $M_0$  write  $C_z = C(M_0 \sqcup 0^z)$  and  $b_z = b(M_0 \sqcup 0^z)$ . On the plane sector  $b_z$  is a Kreweras number, a polynomial in  $z$ . For example  $b(1, -1, 0^z) = \binom{z+2}{2} = 1 + 2z + \binom{z}{2}$ .

**Conjecture 11** (zero insertion). *For every zero-free  $M_0$ , the sequences  $C_z$  and  $C_z - b_z$  have nonnegative forward differences of every order at  $z = 0$ :*

$$C_z - b_z = \sum_{k \geq 0} N_k(M_0) \binom{z}{k}, \quad N_k(M_0) \geq 0.$$

This refines Conjecture 5, since every  $M$  is some  $M_0 \sqcup 0^z$ . It holds on all 10,620 zero-free bases available (`results/zero_insertion.json`), for  $C$  and for  $C - b$ , at every order that fits in  $d \leq 7$ . Where the differences terminate within range,  $C_z$  is a polynomial:

$$C(2, -2, 0^z) = 2 + 5z + 4 \binom{z}{2} + \binom{z}{3}, \quad C(3, -3, 0^z) = 4 + 13z + 16 \binom{z}{2} + 9 \binom{z}{3} + 2 \binom{z}{4}.$$

The degree equals  $\sum_{a>0} a + \#\{a < 0\}$  in all four certified cases. That is too few to conjecture.

## 6 Where a proof has to go

In the geometric model of §2.1, a packet of charge  $p$  reduces by (I3) only to the curvilinear locus  $C_{n,p+1}$  in  $\mathbb{C}^{p+1}$ . The plane case  $p = 1$  is Theorem 2. Conjecture 5 therefore says:

*the integral of  $m_\nu(E)$  over the curvilinear locus in  $\mathbb{C}^{p+1}$  is at least the count of ballot words, which is what the same integral gives in the plane, and exceeds it as soon as  $p \geq 2$ .*

Two routes suggest themselves.

1. *A degeneration.* Degenerate  $C_{n,p+1}$  to a cycle containing a planar component with multiplicity one, plus effective components on which  $m_\nu(\epsilon)$  integrates nonnegatively. Positivity does fail on some individual degeneration components [3], so the decomposition would have to be chosen with care.
2. *A charge-2 Jucys–Murphy calculus* (the naive form is ruled out). Since symmetric functions of the  $J_k$  are central, (I4) says that on the plane

$$\int_{C_{n,2}} f(\epsilon) = \sum_{\lambda \vdash n} \frac{\dim \lambda \chi^\lambda(\sigma)}{n!} f(\text{contents}(\lambda) \setminus \{0\}),$$

a sum over hooks. The exact tables determine the whole functional  $f \mapsto \int_{C_{n,3}} f(\epsilon)$  on symmetric functions of degree  $2d$ : every packet with  $\max M \leq 2$  reduces to  $r = 3$  by (I3). No weights  $w_\lambda$ ,  $\lambda \vdash n$ , make it a content evaluation of the same shape: the linear system (7, 15, 30 equations against 5, 7, 11 unknowns at  $d = 3, 4, 5$ ) is inconsistent (`tools/content_functional.py`). So the curvilinear locus in  $\mathbb{C}^3$  is not governed by any central functional of the Jucys–Murphy elements of  $S_n$ . A charge-2 calculus needs a larger algebra.

3. *A charge-2 Jucys–Murphy calculus (what remains).* Find the analogue, for the curvilinear component of the punctual Hilbert scheme of  $\mathbb{C}^3$ , of the Lehn–Sorger description. In such a calculus the excess  $C(M) - b(M)$  should count a positive family of *non-monotone* or *branched* factorisations.

The exact tables (`results/morin_a{6,7}.json`, 20,725 coefficients) are available to test either route.

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